

CHAPTER :- PLANT KINGDOM

Five Kingdom Classification (Whittaker, 1969): Monera, Protista, Fungi, Animalia, Plantae.

Now excluded from Plantae:

Fungi

Monera with cell wall (e.g., cyanobacteria / blue-green algae)

Protista with cell wall

Present groups under Plantae: Algae, Bryophytes, Pteridophytes, Gymnosperms, Angiosperms.

Classification of Angiosperms – Systems

Artificial Systems

Based on few superficial morphological features (e.g., habit, colour, leaves, androecium structure).

Example: Linnaeus system.

Problem: Separated closely related species; vegetative characters affected by environment.

Natural Systems

Based on natural affinities (external + internal features: anatomy, embryology, phytochemistry, ultrastructure).

Example: Bentham and Hooker.

Phylogenetic Systems

Based on evolutionary relationships, assumes common ancestry.

Modern Approaches:

Numerical taxonomy – uses all observable characters, each given equal weight, analysed by computers.

Cytotaxonomy – chromosome number, structure, behaviour.

Chemotaxonomy – chemical constituents.

ALGAE

Features:

Chlorophyll-bearing, thalloid, autotrophic, mostly aquatic (freshwater + marine).

Also in moist soil, stones, wood, in association with fungi (lichens) or animals (e.g., sloth bear).

Forms: Unicellular, colonial (Volvox), filamentous (Ulothrix, Spirogyra), massive bodies (kelps).

Reproduction:

Vegetative – by fragmentation.

Asexual – by spores (commonly flagellated zoospores).

Sexual – gamete fusion types:

Isogamy – gametes similar (Ulothrix, Spirogyra).

Anisogamy – gametes dissimilar in size (Eudorina).

Oogamy – large non-motile female + small motile male (Fucus, Volvox).

Economic Importance:

Fix ~50% of global CO_2 via photosynthesis.

Increase dissolved O_2 in water.

Primary producers in aquatic food chains.

Food: Porphyra, Laminaria, Sargassum.

Hydrocolloids:

Algin (brown algae)

Carrageenan (red algae)

Agar: from Gelidium, Gracilaria – used in microbial culture, ice-cream, jelly.

Chlorella: protein-rich, used as space food.

Classes of Algae

A. Chlorophyceae (Green Algae)

Features:

Mostly freshwater.

Green due to chlorophyll a & b in chloroplasts (varied shapes: discoid, plate-like, reticulate, cup-shaped, spiral, ribbon).

Pyrenoids: protein + starch storage.

Cell wall: inner cellulose + outer pectose.

Food also as oil droplets.

Reproduction:

Vegetative: fragmentation.

Asexual: flagellated zoospores.

Sexual: isogamy / anisogamy / oogamy.

Examples: Chlamydomonas, Volvox, Ulothrix, Spirogyra, Chara.

Phaeophyceae (Brown Algae)

Habitat: Mostly marine.

Size/Form: Filamentous (Ectocarpus) → giant kelps (up to 100 m).

Pigments: chlorophyll a, c, carotenoids, fucoxanthin (gives brown colour).

Food storage: Laminarin, mannitol.

Cell wall: inner cellulose + outer gelatinous algin coating.

Structure: Holdfast, stipe (stalk), frond (leaf-like).

Reproduction:

Vegetative: fragmentation.

Asexual: biflagellate (unequal flagella) pear-shaped zoospores.

Sexual: isogamy / anisogamy / oogamy. Gametes pear-shaped, biflagellate.

Examples: Ectocarpus, Dictyota, Laminaria, Sargassum, Fucus.

C. Rhodophyceae (Red Algae)

Pigment: Phycoerythrin dominates.

Habitat: Mostly marine, more in warm waters; found near surface & at great depths.

Body: Mostly multicellular; complex thallus.

Food storage: Floridean starch (like amylopectin/glycogen).

Reproduction:

Vegetative: fragmentation.

Asexual: non-motile spores.

Sexual: oogamous, non-motile gametes, complex post-fertilization.

Examples: Polysiphonia, Porphyra, Gracilaria, Gelidium.

Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number and Position of Insertions	Habitat
Chlorophyceae	Green algae	Chlorophyll <i>a, b</i>	Starch	Cellulose	2-8, equal, apical	Fresh water, brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll <i>a, c</i> , fucoxanthin	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral	Fresh water (rare), brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll <i>a, d</i> , phycoerythrin	Floridean starch	Cellulose, pectin and poly sulphate esters	Absent	Fresh water (some), brackish water, salt water (most)

Bryophytes

General Features

Includes *mosses and liverworts*.

Found in moist, humid, shaded areas (often in hills).

Called *amphibians of the plant kingdom*:

Live on soil (terrestrial) but need water for sexual reproduction.

Important in plant succession on bare rocks and soil.

Structure

Plant body more differentiated than algae.

Thallus-like, prostrate or erect.

Attached to substratum by unicellular or multicellular rhizoids.

Lack true roots, stems, and leaves (may have root-like, leaf-like, or stem-like structures).

Main plant body: haploid → gametophyte (photosynthetic, independent).

Sex organs: multicellular

Male: Antheridium → produces biflagellate antherozoids.

Female: Archegonium (flask-shaped) → produces single egg.

Reproduction

Sexual Process:

Antherozoids released into water.

Swim to archegonium; fertilization → zygote.

Zygote → multicellular sporophyte (diploid, attached to gametophyte, dependent on it for nourishment).

Sporophyte: some cells undergo meiosis → haploid spores.

Spores germinate into gametophyte.

Economic & Ecological Importance

Limited direct economic use.

Mosses provide food for some herbivorous mammals, birds, and other animals.

Sphagnum (peat moss):

Yields peat (used as fuel).

Used as packing material for transporting living plants (holds water).

Ecological role:

First colonisers of rocks (along with lichens).

Decompose rocks → form soil for higher plants.

Form dense mats → reduce rain impact and prevent soil erosion.

Classification

Liverworts – e.g., Marchantia

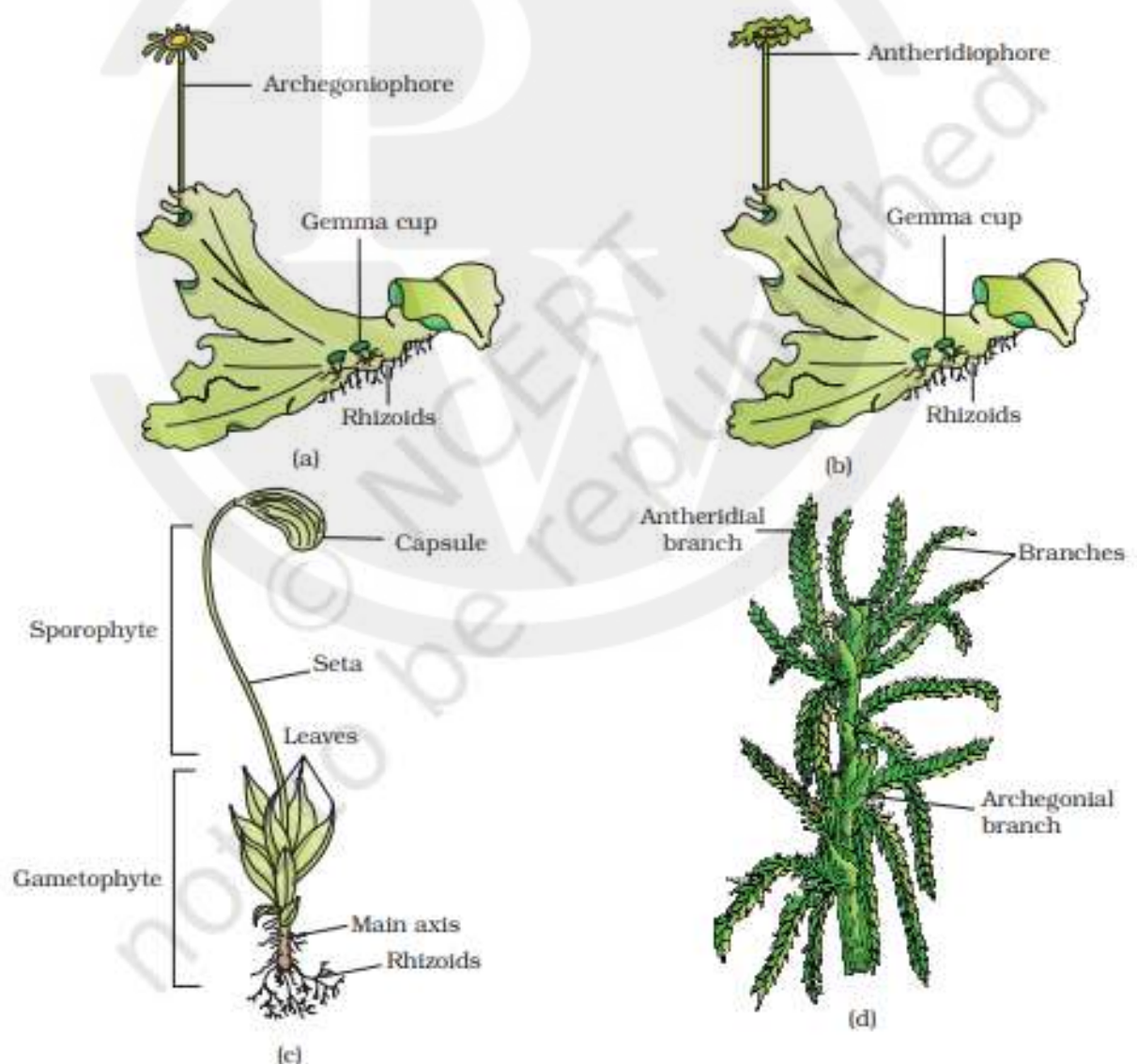
Found in moist, shaded habitats.

Thalloid body; reproduce by fragmentation and gemmae.

Mosses – e.g., Funaria, Sphagnum

Two stages: protonema (creeping) → leafy gametophore.

Sporophyte longer-lived than in liverworts



Liverworts

Habitat

Moist, shady places:

Banks of streams

Marshy ground

Damp soil

Bark of trees

Deep woods

Example: *Marchantia*

Structure

Plant body: thalloid, dorsiventral, closely appressed to substrate.

Leafy members: tiny leaf-like appendages in two rows on stem-like structures.

Reproduction

Asexual

By fragmentation of thallus.

By gemmae:

Green, multicellular, asexual buds.

Form inside gemma cups on thallus.

Detach and germinate into new individuals.

Sexual :- Male and female sex organs on same or different thalli.

Sporophyte differentiated into:

Foot

Seta

Capsule (produces spores after meiosis).

Spores germinate into free-living gametophytes.

Mosses

Life Cycle - Gametophyte Dominant

Two stages:

Protonema stage:

Develops from a spore.

Creeping, green, branched, filamentous.

Leafy stage:

Develops from secondary protonema as lateral bud.

Upright, slender axes with spirally arranged leaves.

Attached to soil by multicellular, branched rhizoids.

Bears sex organs at shoot apex.

Reproduction

Vegetative

By fragmentation.

By budding in secondary protonema.

Sexual

Male organ: Antheridium.

Female organ: Archegonium.

Fertilization → zygote → sporophyte (foot, seta, capsule).

Sporophyte more elaborate than in liverworts.

Capsule produces spores (after meiosis) → dispersed by elaborate mechanisms.

Examples

Funaria, Polytrichum, Sphagnum

Pteridophytes

General Features

Includes horsetails and ferns.

Uses: medicinal, soil-binders, ornamental plants.

First terrestrial plants to have vascular tissues (xylem and phloem).

Habitat: mostly cool, damp, shady places; some grow in sandy soil.

Plant Body

Dominant phase: Sporophyte (unlike bryophytes where gametophyte is dominant).

Sporophyte differentiated into true root, stem, and leaves (all with vascular tissue).

Leaves:

Microphylls – small leaves (Selaginella).

Macrophylls – large leaves (ferns).

Sporophytes bear sporangia on sporophylls (leaf-like appendages).

Sporophylls may form strobili/cones (Selaginella, Equisetum).

Reproduction

Sporophyte → Gametophyte

Sporangia → spores (by meiosis in spore mother cells).

Spores germinate into prothallus:

Spores germinate into prothallus:

Small, multicellular, thalloid.

Free-living, mostly photosynthetic.

Needs cool, damp, shady environment.

Gametophyte Stage

Bears antheridia (male) and archegonia (female).

Fertilization requires water for antherozoid movement.

Zygote → multicellular sporophyte (dominant phase).

Spore Types

Homosporous: one type of spore (majority of pteridophytes).

Heterosporous: two types of spores –

Megaspores (large, female gametophytes).

Microspores (small, male gametophytes).

Examples: *Selaginella*, *Salvinia*.

In heterosporous plants:

Female gametophyte retained on parent sporophyte.

Zygote develops into embryo inside female gametophyte.

Precursor to seed habit.

Classification

Psilopsida – *Psilotum*

Lycopsida – *Selaginella*, *Lycopodium*

Sphenopsida – *Equisetum*

Pteropsida – *Dryopteris*, *Pteris*, *Adiantum*



Figure 3.3 Pteridophytes : (a) *Selaginella* (b) *Equisetum* (c) Fern (d) *Salvinia*

Gymnosperms

Gymnos = *naked*, sperma = *seeds*.

Ovules not enclosed by ovary wall → remain exposed before and after fertilisation.

Seeds formed post-fertilisation are naked.

General Features

Medium-sized to tall trees, some shrubs.

Example: Sequoia – tallest tree species.

Roots:

Tap roots (common).

Mycorrhizal roots (Pinus) – fungal association.

Coralloid roots (Cycas) – with N_2 -fixing cyanobacteria.

Stem:

Unbranched (Cycas).

Branched (Pinus, Cedrus).

Leaves:

Simple or compound.

Adaptations: thick cuticle, sunken stomata, needle-like leaves (conifers) to reduce water loss.

Reproduction

Heterosporous: produce microspores (male) & megaspores (female).

Male cone (microsporangiate strobilus):

Microsporophylls with microsporangia → microspores → pollen grains (reduced male gametophyte).

Female cone (megasporangiate strobilus):

Megasporophylls with ovules (megasporangia).

Ovule = nucellus + integuments.

Megaspore mother cell → meiosis → 4 megaspores → 1 develops into female gametophyte with 2+ archegonia.

Gametophytes:

Not free-living (remain within sporangia on sporophyte).

Fertilisation:

Pollen grains → carried by wind → reach ovule opening.

Pollen tube grows to archegonia → releases male gametes.

Zygote → embryo; ovule → naked seed.

Reproduction

Heterosporous: produce microspores (male) & megaspores (female).

Male cone (microsporangiate strobilus):

Microsporophylls with microsporangia → microspores → pollen grains (reduced male gametophyte).

Female cone (megasporangiate strobilus):

Megasporophylls with ovules (megasporangia).

Ovule = nucellus + integuments.

Megaspore mother cell → meiosis → 4 megaspores → 1 develops into female gametophyte with 2+ archegonia.

Gametophytes:

Not free-living (remain within sporangia on sporophyte).

Fertilisation:

Pollen grains → carried by wind → reach ovule opening.

Pollen tube grows to archegonia → releases male gametes.

Zygote → embryo; ovule → naked seed.

Angiosperms

Ovules enclosed in ovary.

Seeds enclosed in fruits.

Also called flowering plants.

General Features

Largest group of plants; found in diverse habitats.

Size range: smallest Wolffia to tallest Eucalyptus (>100 m).

Economic importance: food, fodder, fuel, medicines, timber, etc.

Reproduction

Flowers produce pollen grains & ovules in specialised structures.

After fertilisation → ovary develops into fruit enclosing seeds.

Classification of Angiosperms

Angiosperms (flowering plants) are broadly divided into two main classes based on the number of cotyledons in the seed and several other characteristics:

1. Dicotyledons (Dicots)

Key features:

Cotyledons – The seed contains two cotyledons.

Venation – Leaves usually have reticulate (net-like) venation.

Root system – Usually a tap root system.

Floral parts – Generally arranged in multiples of 4 or 5.

Vascular bundles – Arranged in a ring in the stem; open type (with cambium, capable of secondary growth).

Examples – Pea, Rose, Sunflower, Mustard, Mango.

2. Monocotyledons (Monocots)

Key features:

Cotyledons – The seed contains only one cotyledon.

Venation – Leaves usually have parallel venation.

Root system – Usually a fibrous root system.

Floral parts – Generally arranged in multiples of 3.

Vascular bundles – Scattered in the stem; closed type (without cambium, no secondary growth).

Examples – Wheat, Rice, Maize, Banana, Sugarcane, Lily.